Practice workbook · Integrating trig functions using identities

Rewriting trig products or powers as sums of sin(ax) and cos(ax) with identities, then integrating and evaluating exactly.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-17

1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

6 (a) Show that

$$\sin(A + B) + \sin(A - B) = 2\sin A \cos B$$

(2)

(b) Hence express $2\sin 7x \cos x$ in the form $\sin mx + \sin nx$ where m and n are integers, giving the value of m and the value of n.

(1)

(c) Use calculus to evaluate

$$\int_0^{\frac{\pi}{4}} (6\sin 7x \cos x)\,dx$$

(4)

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1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

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Question 6 continued

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1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

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Question 6 continued

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1. Question 6 · part c · 4 marks

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Worked-example retry · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 7 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part whole only.

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1 Using calculus find the exact value of $\int_{\frac{\pi}{4}}^{3} \cos 4\theta \, d\theta$

Give your answer in the form $-\frac{\sqrt{a}}{b}$ where a is a prime number and b is an integer.

(4)

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(Total for Question 1 is 4 marks)

2. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

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3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 15 marks)

3. Question 10 · part d · 4 marks · Working and answer

Working and answer

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4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P + Q}{2}\cos\frac{P - Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x \cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx$$

(4)

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4. Question 10 · part e · 4 marks

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Further practice · Attempt part e only.

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Question 10 continued

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4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

Question 10 continued

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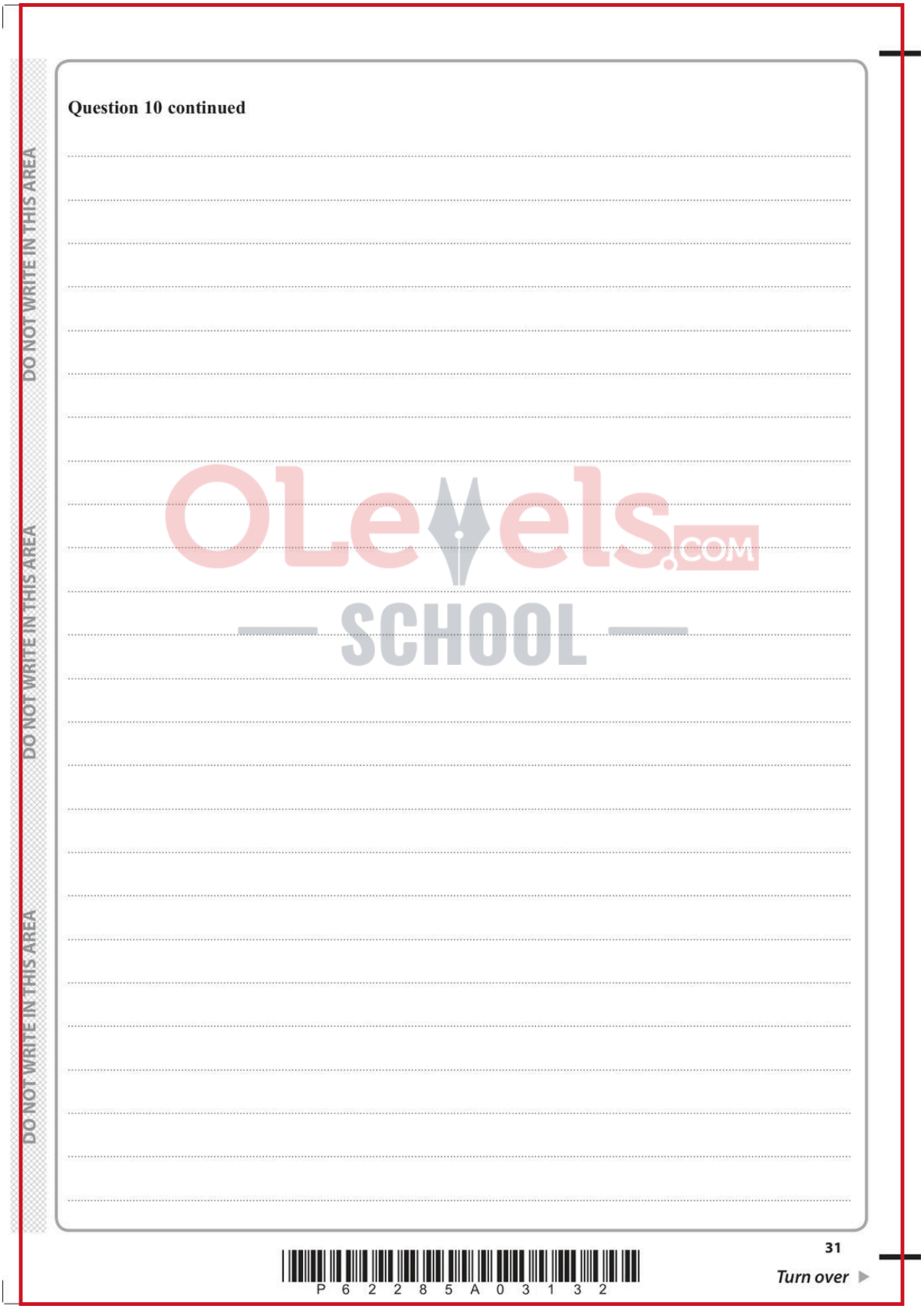
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4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.



Question 10 continued

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Turn over ▶

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

Question 10 continued

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(Total for Question 10 is 16 marks)

4. Question 10 · part e · 4 marks · Working and answer

Working and answer

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5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

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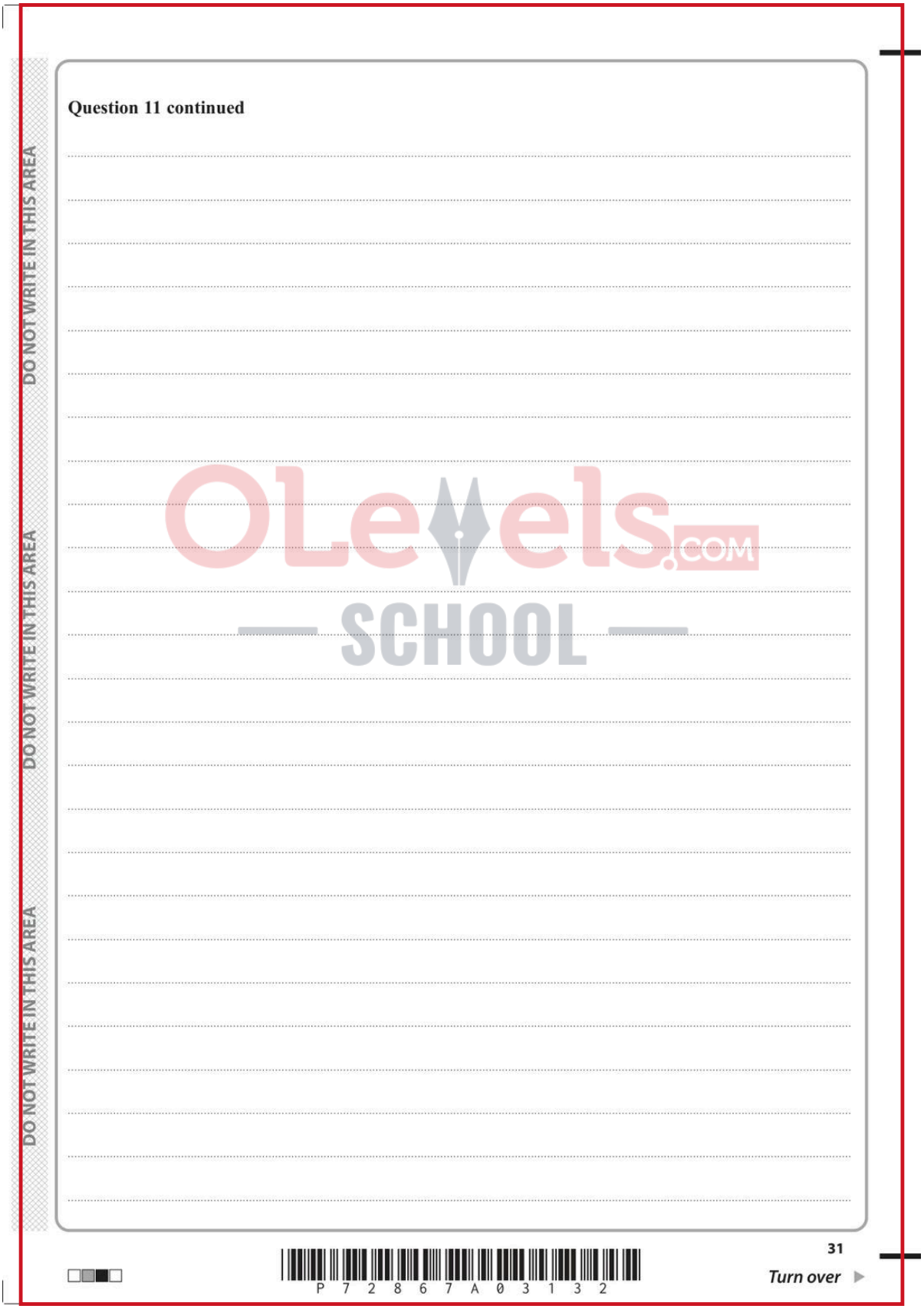
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5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.



5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 15 marks)

5. Question 11 · part c · 4 marks · Working and answer

Working and answer

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6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$

(3)

(ii) $\sin 2A = 2\sin A\cos A$

(1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$

(4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

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6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 16 marks)

6. Question 11 · part d · 4 marks · Working and answer

Working and answer

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7. Question 9 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

Diagram **NOT** accurately drawn

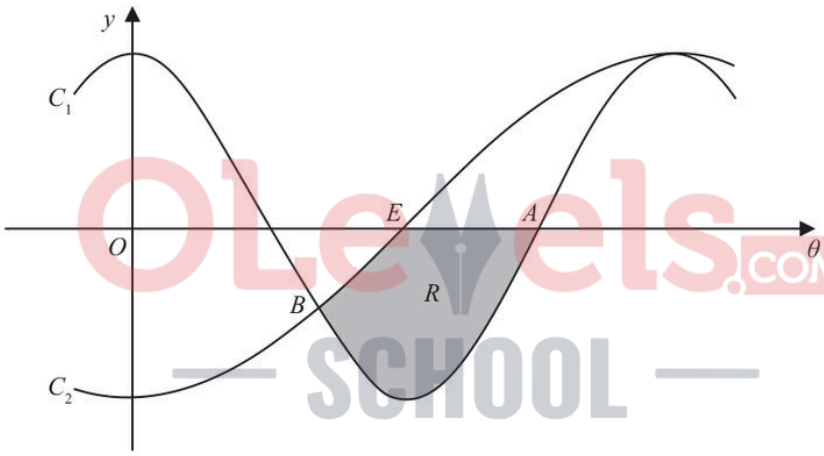


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

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7. Question 9 · part b · 4 marks · Working and answer

Working and answer

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8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 Using formulae from page 2

(a) show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

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8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 16 marks)

8. Question 10 · part d · 6 marks · Working and answer

Working and answer

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9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

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9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

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Question 10 continued

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9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

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Question 10 continued

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9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

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Question 10 continued

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(Total for Question 10 is 17 marks)

9. Question 10 · part d · 4 marks · Working and answer

Working and answer

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